Opt-in exact linear-root policy

Run **linear-root-20261026-v1**, 2026-10-26. Intel Xeon Platinum 8573C, managed Linux x86_64, L0. Public synthetic/known-answer fixtures only. No admitted wall time.

Frozen task and accounting

All ten original curves, B=8, arities 2 and 3, target label 0. Twenty configurations and 200 requested processes: direct A/A, direct, helpers and invariants with each policy, plus group A/A and MITM.

The same cold finite-decomposition interval is charged for policy off and on, including actual construction, F4, roots, original minimum-mask sign search, infinity exceptions, verification, deduplication and serialization. Limits: 15,000 ms cooperative, 18-second hard timeout plus one-second grace, 4 GiB, 32 MiB/file; 3,800 seconds/1 GiB globally. F4_FP_LINEAR_ROOTS=1 prefers an analytically solved pure linear basis equation before the generic full-field root scan; default remains off.

Protocol · source, command and transport receipt · independent native audit · raw records.

The receipt records the source revision, pinned binary hashes and exact command where captured. All workers run on already observed training targets; final E28 labels 32–63 remain unopened.

Recorded observations

All 200 outputs match the entire frozen signed-row list. Independent PARI verifies 980 positive sums and 980 wrong-sign controls. Policy-on/off median ratios are direct 0.9994338578968132, helpers 0.9992032880251398 and invariants 0.996520508243923. Best helper cell ratio is 0.8421707962912168; regressions are retained. All three polynomial variants still exceed ordinary group MITM in every one of twenty matched cases. Two new unit tests cover exhaustive small-field linear coefficients, rejected non-pivots and recursive/inconsistent differential answers without mutating test-process environment.

Arm	Requested	Accepted
Direct A/A	20	20
direct-aa + linear roots	20	20
Direct F4	20	20
Direct + linear roots	20	20
Pair invariants	20	20
Invariants + linear roots	20	20
Group A/A	20	20
Group MITM	20	20
Eight helpers	20	20
Helpers + linear roots	20	20

Completed and independently accepted workers

linear-root-20261026-v1; unit: process cells; all requested arms retained

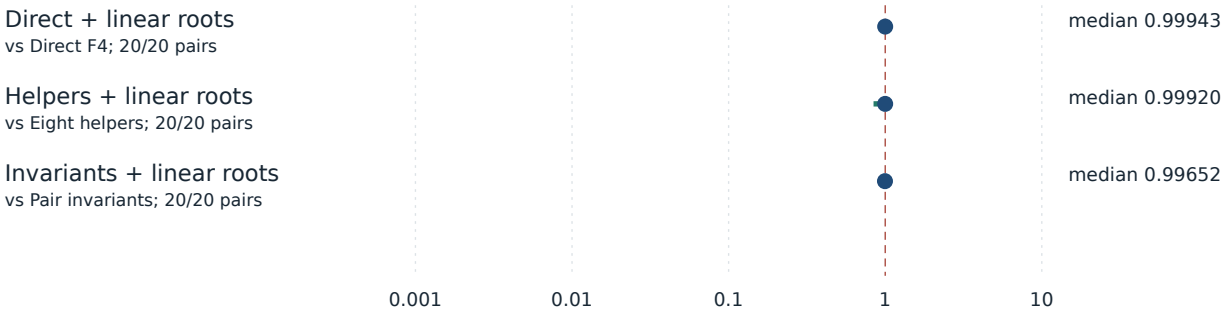


Green: accepted. Gray: remaining requested cells. Unit: process cells.

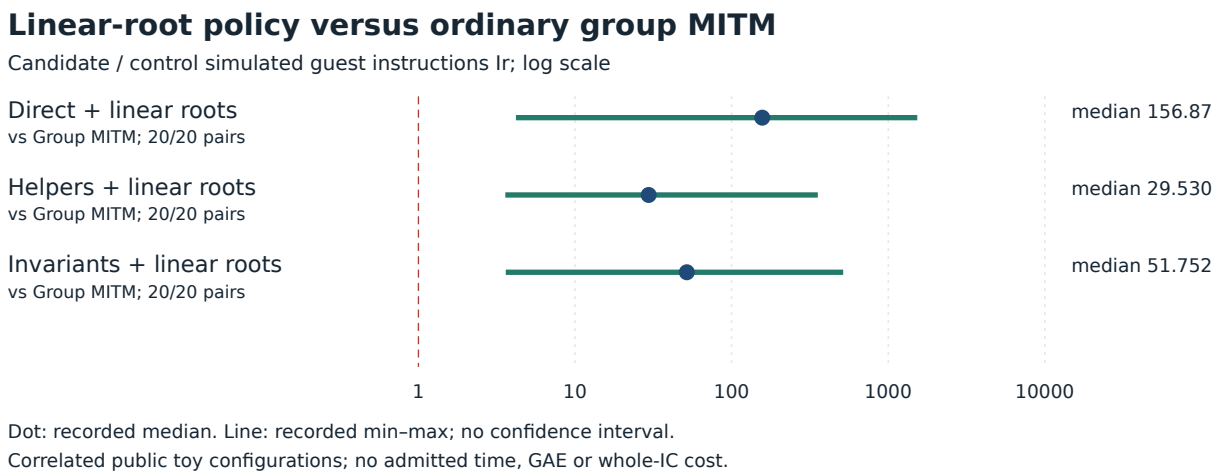
Candidate / control	Pairs	Fewer Ir	Median ratio
Eight helpers / Direct F4	20/20	20	0.59112
Pair invariants / Direct F4	20/20	20	0.63021
Group MITM / Direct F4	20/20	20	0.11377
Direct + linear roots / Direct F4	20/20	16	0.99943
Helpers + linear roots / Direct F4	20/20	20	0.58332
Invariants + linear roots / Direct F4	20/20	20	0.62117
Direct + linear roots / Group MITM	20/20	0	156.87
Helpers + linear roots / Eight helpers	20/20	11	0.99920
Helpers + linear roots / Group MITM	20/20	0	29.530
Invariants + linear roots / Pair invariants	20/20	17	0.99652
Invariants + linear roots / Group MITM	20/20	0	51.752

Completed paired instruction ratios

Candidate / control simulated guest instructions Ir; log scale



Dot: recorded median. Line: recorded min–max; no confidence interval.
Correlated public toy configurations; no admitted time, GAE or whole-IC cost.



Validation, limits and raw preservation

This is a generic implementation optimization, not new twist geometry. A task/env assertion and independent audit verify the selected policy. Nonlinear root scans remain unchanged. No external full-root reference, complete IC cost, matched rho, scaling or fastest implementation result is supplied. The pre-feature library check is retained separately; final checks rerun after this shared solver change.

Independent PARI positive and negative checks each: **980**. The native gate reopens script, stdout, stderr and actual-exit hashes; only the documented maximum-stack warning is allowed.

Measured instructions are simulated guest instructions Ir, not CPU cycles, seconds, field operations or GAE. Ratios are native-audited candidate/control values. Counts repeat correlated configurations; A/A repeats check only the frozen five-percent within-input gate. No confidence interval or independent-host replication is claimed. Missing pairs stay missing. No matched rho or validation-gated benchmark row is added.

Raw bytes, failed/capped cells and loose replay metadata are preserved by the deterministic archive manifest. The user-selected remote is **s3://setec-astronomy-artifacts/store**, logical name is recorded in the manifest.

Archive SHA-256:

`f0b73ee2cd1ab71cec520098a6c4e4868a0a0846daaf64c207be6fcf0839be7a`

Extraction/member check instructions.

Editable SVGs and recorded plot data accompany this report. No result is extrapolated to NIST-size DLPs, full index calculus or novelty.